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# Experimental Investigation of a Multi Tube Burner for Premixed Hydrogen and Natural Gas Low Emission Combustion

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*Hydrogen as an essential part of future decarbonization of the energy industry makes it a crucial necessity to replace conventional, natural gas based concepts in gas turbine combustion. This paper presents an experimental study of a multi-tube jet flame burner. The study is carried out with natural gas and pure hydrogen fuel at gas turbine relevant conditions at atmospheric pressure. To identify key differences between hydrogen-air and natural gas–air flames on the overall robustness and flame flashback behavior, air bulk velocity (80–120 m/s), adiabatic flame temperature (1235–2089 K) and air inlet temperature (623–673 K) are varied over a wide range, covering a range of Reynolds numbers of 10,000–20,000. Depending on flame temperature, two different flame shapes are observed for natural gas–air flames. The shape of the hydrogen-air flame changes less over the range of flame temperatures tested, but is generally more compact. The process of fuel-air mixing is further investigated by concentration distribution measurements in a water tunnel setup. Therefore, planar laser-induced fluorescence is utilized for visualization. The measured concentration distributions confirm the overall good mixing quality but also give an explanation on the observed flashback behavior of the different burner designs at reacting tests. The findings of the study are composed in a flashback correlation combining the observed flashback drivers for the burner configurations investigated.*
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*Keywords: hydrogen combustion, jet flame, multi tube, flashback, gas turbine*

## Introduction

The combustion of hydrogen in gas turbines is a viable pathway for the re-electrification of hydrogen generated by electrolysis with the help of surplus wind or solar power. Especially, gas turbines employed for the generation of elevated temperature steam and electrical power at the same time as well as compressor drives at remote locations are difficult to substitute with electrical steam generators or drives, respectively.

While fuel flexible diffusion gas turbine combustors are capable of burning hydrogen ($H_2$) and natural gas (NG) in the full range, they also produce excessive $NO_x$ (nitrogen oxides) emissions [1]. The challenges to operate dry ultralow emission combustion systems with hydrogen and natural gas at the same time are much higher than traditional diffusion combustors because premixed combustion tends to promote thermoacoustic instabilities [2]. Additionally, fuel-flexible premixed combustors struggle to ensure flame stability at part load conditions with natural gas and to provide high flashback (FB) safety at base load conditions for hydrogen while at the same time producing ultralow $NO_x$ emissions.

Recently, advances have been reported for commercially available industrial gas turbines that feature fuel flexibility from pure natural gas to higher hydrogen blending ratios in the range of 40–60 vol. % hydrogen with only small changes in the combustor design [3–5]. Here, the key challenge is to remain in a low emission mode without the risk of flame flashback. However, in order to achieve significant $CO_2$ saving in the range of 50% and more, volumetric hydrogen contents of 80–100 vol. % in the fuel are needed because of the extreme difference in density between hydrogen and natural gas. Currently, there are commercially available systems, e.g., based on micromix technology. Since they are based on small diffusion flames with only limited premixing, they feature lower but not ultralow $NO_x$ emissions [6]. At the same time, these systems suffer of limited fuel flexibility in terms of combustion of pure natural gas [7]. This is because the flame stabilization of micromix based combustors relies inherently on the high reactivity of the hydrogen fuel. Natural gas flames, however, have much lower flame speeds, especially, at gas turbine part load and the associated lower flame temperatures. Another industrial system capable of hydrogen that relies on a trapped vortex to stabilize the flame has been presented recently by Koomen et al. [8].

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It features high fuel flexibility at low $NO_x$ emissions but is currently only validated for a maximum pressure of 6.9 bar and 280 °C inlet temperature. Thus, the currently unsolved challenge is to avoid flame flashback at base load flame temperatures and high pressure with pure hydrogen and at the same time avoid lean blow out (LBO) at part load with natural gas, while remaining in the regime of low emissions and thermoacoustically stable combustion.

A promising approach to handle these diverging demands is a burner design with arrays of premixed jet flames. Lammel et al. [9] reported successful experimental high pressure tests with very high hydrogen contents using a multi-tube combustor. The combustor features 12 relatively large tubes circumferentially arranged in the combustor. In contrast to that, York et al. [10] developed a multi-tube burner with a very large number of smaller jets. Multiple of these burners were used to achieve the full engine power of a single can. They pointed out the potential for hydrogen combustion but the study was focused on low emission combustion of natural gas at elevated turbine inlet temperatures. A similar approach has been presented by Asai et al. for hydrogen rich fuels [11] but with a convex shaped mixing passage. Witzel et al. have recently reported that they have conducted high pressure combustion tests with natural gas and pure hydrogen at gas turbine relevant conditions. They state to use jet burner technology but do not provide details on the investigated geometry [12]. Besides these industrial systems, there is a large number of general investigations for academic single jet flames like, for instance, Refs. [13] and [14].

Studies conducted by Kalantari et al. [15] on flame flashback with hydrogen fuel showed experimentally that gas turbine relevant conditions like increased combustor pressure, high flame temperatures, and high compressor outlet temperatures promote flashback. The increased chemical reactivity at elevated temperatures leads to turbulent flame speeds locally exceeding the convective flow velocities of unburnt premixed gas. Especially at the boundary layers of the combustor walls, the flow velocities are locally small enough to facilitate flashback [16,17]. With the onset of flashback, a separation zone becomes present upstream of the anchored flame [18]. In this backflow region, ignition of unburnt gas is governed by heat transfer from the flame front to the external streamlines and the wall. At constant air mass flow, an increase in the height of the backflow region is observed as the equivalence ratio increases [15]. When this region grows large enough to overcome the quenching effect of the heat loss, flashback occurs. Therefore, boundary layer flashback is the most likely type of flashback related to jet flames at high mixing tube velocities. From a general point of view, jet burners are a promising approach to handle the challenges regarding premixed hydrogen flames because the pressure loss of the burner, which is strictly limited from a gas turbine efficiency point of view, is converted almost fully into axial velocity, which protects the burner most efficiently from wall induced flashback. High axial velocities in the range of 55–85 m/s seem to be sufficient to avoid wall induced flashback by hydrogen for gas turbine relevant conditions in an ideal flow channel [15] and can be realized by jet burners with a pressure loss in the range of 1.5% of the combustor pressure. In contrast to that, traditional swirl combustor concepts offer by design many zones of small axial velocity at the walls and in the center of the swirl vortex to allow for a stable operating of natural gas. They demand much higher pressure losses to achieve sufficiently high axial bulk velocities for flashback safe hydrogen combustion.

This paper investigates two multi-tube premixed jet burner concepts with different fuel gas injectors for pure natural gas and pure hydrogen fuel. The focus of the investigation is the atmospheric assessment of the premixed operating regime of the MAN MGT 6000 industrial gas turbine. However, the investigated combustors are only capable to deliver a portion of the full thermal power necessary at full load and full pressure conditions. Furthermore, only single stage operation is considered in this study. For the first time, to the knowledge of the authors, a practically relevant multi-tube premixed burner with helical tubes is experimentally investigated for pure hydrogen and natural gas in terms of flashback, mixing quality and emissions. We call this concept helix burner. The burner is combined with different fuel gas injector concepts and it is compared with a more traditionally designed jet burner with straight tubes.

The remainder of this paper is structured as follows: First the experimental setup and the tested operating points are described, subsequently, the results of the water channel measurements are discussed as well as the combustion experiments. Finally, the results are summarized, and conclusions are presented.

## Experimental Setup and Diagnostics

**Burner Setup.** A schematic sketch of the test rig is provided in Fig. 1. Two geometrical variants of the multi-tube burner are employed in the experiments: the jet burner and the helix burner. Each burner consists of 19 individual mixing tubes arranged in an inner and outer ring of 6 and 13 tubes, respectively. The diameter of all tubes is equal. They end in the base plate, also referred to as burner plate, which separates the burner plenum from the combustion chamber. While the straight mixing tubes of the jet burner form an orthogonal outlet in the burner plate, the twisted shape of the helix burner results in an angled outlet, seen in Fig. 1 on the right, from which a swirling motion of the flow results. The burners are tested with three different types of fuel injectors, referred to as nozzles. A sketch of the nozzles is provided in Fig. 1 on the left.

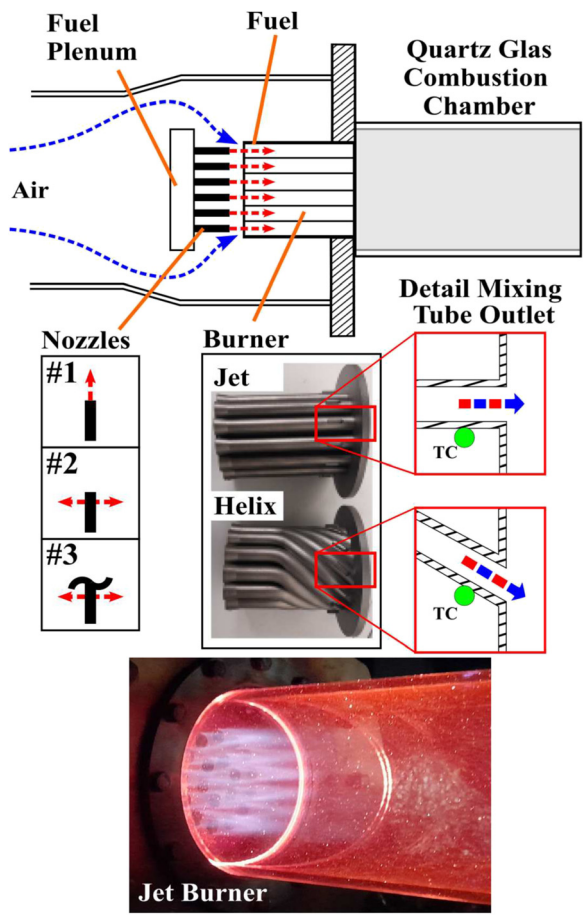


**Fig. 1 Schematic of the burner setup (top) with preheat air flow (blue dashed lines) and fuel injection (red dashed lines). In the middle on the left side, the different nozzle configurations are sketched to illustrate their working principles. On the right side in this row, the burners are shown with a detailed sketch of the mixing tube outlet into the combustion chamber. The green dots mark the position of the thermocouples (TC) plugged into the sockets (visible in the middle photo of the burners). The bottom row shows a visual impression of the jet burner firing hydrogen. (Color version online.)**



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They can be differentiated according to their degree of complexity in terms of fuel–air interaction at the injection point. Configuration #1 is the most simple, where the fuel jet is injected coaxially into the air. Nozzle #2 and #3 are jet-in-crossflow configurations, whereas #3 is additionally equipped with turbulators near the fuel injection hole to enhance the mixing. The different injectors are not mixed within one single configuration. Hence, six experimental configurations consisting of two burner geometries in combination with three fuel gas injection variants are investigated. The basic mode of operation remains the same for all burner configurations.

Preheated air enters the burner from the left-hand side (illustrated at the top in Fig. 1, indicated by the blue dashed line) and flows around the fuel plenum. When reaching the front section of the burner, the air flow splits homogeneously into the 19 mixing tubes. At this location, the fuel is injected into the air stream and mixes on the way through the tubes, before entering the combustion chamber at the burner plate. Hence, fuel and air are technically premixed. The mixing length is constant for both burner geometries and in order to provide optical access to the flame, the cylindrical combustion chamber consists of quartz glass. To achieve a more symmetric fuel distribution for the nozzles, the main fuel line, which is controlled by a coriolis type flowmeter, is split into two identical lines before entering the fuel plenum at opposite points. From the fuel plenum, the fuel is injected through a nozzle in each mixing tube. Consequently, the fuel amount going through each nozzle is only determined by their individual pressure loss. To ensure a homogeneous distribution of the fuel amongst all nozzles, the individual pressure losses have been measured beforehand, and outliers have been sorted out. Therefore, the nozzles utilized have an overall standard deviation of less than 2.75% of the measured averaged pressure losses.

To detect a possible flashback event, each mixing tube is equipped with a thermocouple that tracks the individual tube material temperature shortly upstream of the burner plate. Small sockets are embedded in the mixing tube outer surface (see Fig. 1) to hold each thermocouple in place and to ensure that there is no convective influence from the surrounding burner volume. In this study, a flashback is defined as a flame anchoring inside of at least one of the mixing tubes. With the flame inside a mixing tube, the measured temperature increases rapidly within very few seconds, which makes the FB unfailingly detectable. Such a flashback detection technique was successfully employed in other works [15,17]. The combination with a pressure probe for a faster detection was also tested. Therefore, a pressure probe was installed inside the burner plenum near the burner inlet. However, the flashback of a single mixing tube results in only a very small pressure rise in the burner plenum, which cannot be reliably detected by the sensor due to the low signal-to-noise ratio. Additionally, the temperature sensors are used to evaluate the base plate temperature of the burners.

Flame shape and position are determined by line-of-sight OH*–chemiluminescence images, since the OH* radical qualitatively correlates with the location of the heat release zone which indicates the flame front. Flame detection is managed by a bandpass filtered, intensified high-speed camera with a center wavelength at $\lambda = 310$ nm recording at 10,000 fps, an exposure time of 1/fps, and a spatial resolution of 3.5 px/mm. The acquisition time of 0.728 s leads to an averaging over 7280 frames for the mean OH* intensity images presented in this study. In addition, the time resolved flame images can provide a visual confirmation for flashback.

Exhaust gas for the emission measurement is extracted in the exhaust tube with a gas probe installed 1080 mm downstream of the burner plate. The gas sample is guided to the gas analyzer by a heated tube to prevent condensation and is analyzed for $NO_x$ on a wet basis by a chemiluminescence detector. For more details on the emission analyzing process refer to [19]. Behind the gas probe, at the end of the exhaust tube, an orifice with an area contraction ratio of 0.23 is installed to suppress thermoacoustic instabilities. The influence of area contraction on the combustion process was investigated by Terhaar et al. [20] and is therefore not expected to have an influence on the combustion process.

**Water Tunnel.** The reliable operation of combustion devices depends to a large degree on the mixing process [21]. A good measure for the mixing quality of the main air flow and the fuel is the fuel concentration distribution, but visualization of natural gas in air is challenging and quite harder for hydrogen. To qualitatively assess the resulting mixing quality of the different burner configurations, planar laser-induced fluorescence (PLIF) measurements are conducted in a water tunnel. With the law of similarity, the flow conditions that determine the mixing process are transferred from air to water. For this purpose, the Reynolds number and impulse ratio of the fuel surrogate are adjusted to correspond to the values set in the combustion tests, which allows the evaluation of the mixture quality using the water tunnel. Water with dissolved Rhodamin 6G dye is utilized as the fuel surrogate with a fluorescence peak at $\lambda = 480$ nm. For the dye excitation, a CW-Laser (continuous wave) with a maximum power output of 4200 mW at a wavelength of $\lambda = 532$ nm is applied. The laser illuminates a horizontal plane of 1 mm thickness at a position 3 mm downstream from the exit of the mixing tubes. Image sampling of the emitted fluorescence is accomplished by a bandpass filtered, intensified CMOS camera with a recording frequency of 8 Hz, and a spatial resolution of 4 px/mm.

The water tunnel and the setup for the experiments are shown in Fig. 2. For more details about the water tunnel test rig the reader is referred to Ref. [22]. The time-averaged fluorescence light intensity $\langle I(x) \rangle$ is related to a known homogeneous intensity distribution $I_{\mathrm{hom}}(x)$, both corrected for background intensity $I_{\mathrm{bg}}(x)$. Subsequently, the respective global mean concentration $\bar{C}$ is subtracted to obtain the time-averaged concentration fluctuation $\langle \tilde{C}(x) \rangle$, defined as

$$\left\langle \tilde{C}(x) \right\rangle = \frac{\langle I(x) \rangle - I_{\mathrm{bg}}(x)}{I_{\mathrm{hom}}(x) - I_{\mathrm{bg}}(x)} - \bar{C} \tag{1}$$

To evaluate the temporal mixing quality and the level of intermittency, the intensity of segregation as proposed by Danckwerts [23] or in a more common formulation, the unmixedness $U_t$ is calculated with

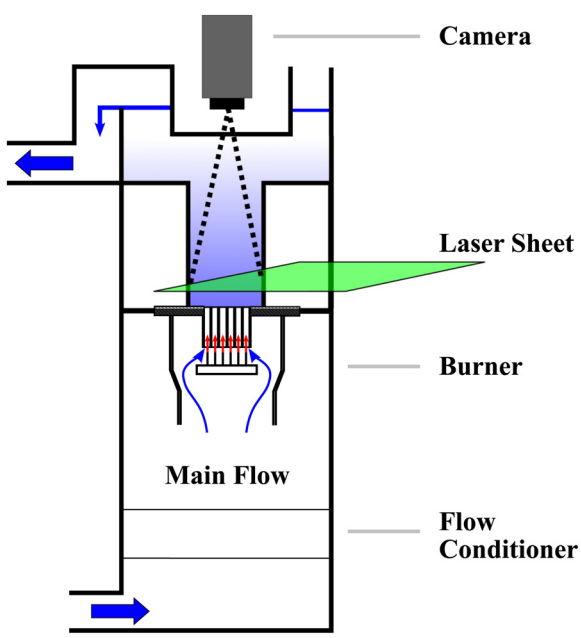


**Fig. 2 Water tunnel setup for PLIF measurements. Flow direction is bottom to top, with the measurement plane (laser sheet) 3 mm downstream of the base plate in the combustion chamber.**



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$$U_t = \frac{\sigma_C^2}{\overline{C(x)}\left(C_{\max} - \overline{C(x)}\right)} \tag{2}$$

The utilized parameter, in our case the measured intensity, is determined at each position of the image plane. At a point of maximum variance of the flow, with $\overline{C(x)}(C_{\max} - \overline{C(x)}) \rightarrow \sigma_C^2$, it becomes unity. This corresponds to equal shares of pure air and pure fuel passing the image plane. A value of 0 is reached for perfectly mixed conditions.

**Tested Conditions.** This section gives a brief overview of the tested conditions at the atmospheric test rig and at the water tunnel. The test conditions for the multi-tube burner are down-scaled from full load gas turbine operating conditions. Furthermore, only single stage operating is considered in this study. Thus, the covered thermal power is in the range of 50–100 kW with adiabatic flame temperatures of up to 2100 K. The investigated preheat temperatures (623 K and 673 K) are chosen to simulate the gas turbine compressor exit temperatures relevant for gas turbine combustion. Pure hydrogen and pure natural gas are used as fuel. Hydrogen is supplied by pressurized bottles, while natural gas is supplied by the pipeline grid of Berlin. In case of hydrogen the whole operating window, from lean blow-off to flashback is investigated. Depending on the air mass flow, the mixing tube bulk velocity ranges from 80 to 120 m/s, which results in a range of Reynolds numbers of 10,000–20,000 with regard to a single tube. The fuel momentum ratio $J$ for the conducted nozzles ranges from 0.65 to 28 and 0.5 to 19 for hydrogen and natural gas, respectively. An overview of the fuel momentum ratios of the different nozzle variants is provided in Fig. 3. A higher momentum ratio is achieved for hydrogen at the same adiabatic flame temperature compared to natural gas. Due to the design of the combustor, the fuel is preheated on its way to the injection point to an assumed temperature of 523 K. This calculated fuel temperature and a measured air temperature of 673 K are used to calculate the similarity for the water tunnel experiments. The transfer of flow conditions from gas to water results in a reduction of the mixing tube bulk velocity to a range of 0.8–1.3 m/s, since the same combustor geometry is used as for reactive conditions. The mass flow ratio of the injected fuel surrogate and hydrogen fuel is $\dot{m}_{\text{water}}/\dot{m}_{\text{hydrogen}} = 74.3$.

With the measured fuel and air mass flows at a given preheat temperature, the laminar flame speed and the adiabatic flame temperature can be calculated with the chemical kinetic solver CANTERA and the GRI 3.0 mechanism. The associated turbulent combustion regime of the multi-tube burner is depicted in Fig. 4. Both, the turbulence intensity $v' = 0.227\,\text{Re}^{-1/10}$ [25] and the turbulent integral length scale $l_{\text{int}}$ are characterized by the mean flow properties of the jet flames. A constant value of 20% of the mixing tube diameter is chosen for $l_{\text{int}}$, which is in agreement with the value observed by Aleyasin et al. [26] for the present flow conditions. The thermal flame thickness $\delta_L = (T_b - T_u)/(dT/dz)_{\max}$ [27] is calculated based on the maximum temperature gradient, whereby the subscripts $b$ and $u$ denote burned and unburned conditions, respectively. Most of the data points are within the thin reaction zone, which represents the common practical operating range of gas turbines. Since the bulk velocity for hydrogen changes only slightly compared to the sharp increase in laminar flame speed, the data points in Fig. 4 transition to the wrinkled flamelets regime with increasing thermal load. Consequently, the chemical reactivity of hydrogen predominates and the interaction between flame front and the turbulent flow field is assumed to become kinematic in terms of the interaction between flame front and turbulence. All observed FB events occurred in the flamelet regimes and thus lie outside the thin reaction zones regime. In the case of hydrogen, the LBO points are also not within the thin reaction zone regime but in the distributed reactions regime.

## Results and Discussion

**Isothermal Experiments—Mixing Quality.** Fuel-air mixing and the resulting quality of fuel concentration distribution prior to combustion is essential for various aspects of gas turbine operation. Primarily it affects $NO_x$ emissions but also (thermoacoustic) combustion stability. Qualitative concentration distribution measurements are conducted in the water tunnel by means of PLIF. The mixing capability of the helix and the jet burner is compared in Fig. 5, showing the fluctuation around the mean concentration distribution. The burners are equipped with nozzle #1 and the momentum ratio varies in the range of $J = 3$–28. These conditions correspond to adiabatic flame temperatures of $T_{\text{ad}} = 1235$–2089 K at an air mass flowrate of $0.8\,\dot{m}_{\text{nom}}$ in the reactive tests. The water tunnel study focuses on nozzle #1 in combination with the different characteristics of the two burner geometries. Nozzle #1 is employed because its flow interaction efficiently reduces FB propensity, which will be discussed later. However, the mixing quality increases with the complexity of the nozzle–flow interaction from nozzle #1 to nozzle #3 (results not shown).

For both tube rings of the jet burner (Fig. 5, first row), it is observed that the fuel-rich share of the jets is attached to the outer

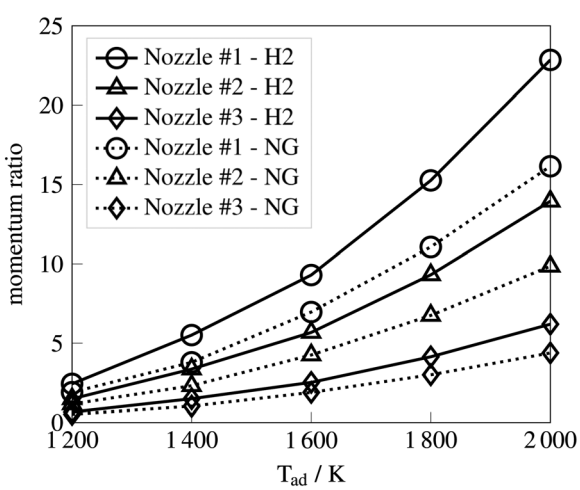


**Fig. 3 Fuel momentum ratios for the investigated range of adiabatic flame temperature for hydrogen (solid line) and natural gas (dashed line). The fuel momentum ratio depends on the nozzle, since the number of holes and the diameter of the holes change.**

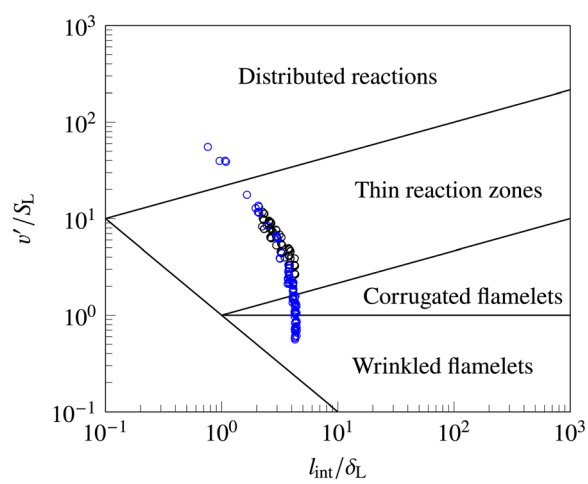


**Fig. 4 Turbulent combustion regimes for the multi-tube burner at the tested conditions. Natural gas is shown with black symbols and hydrogen with blue symbols (flame regimes defined by Peters [24]). (Color version online.)**



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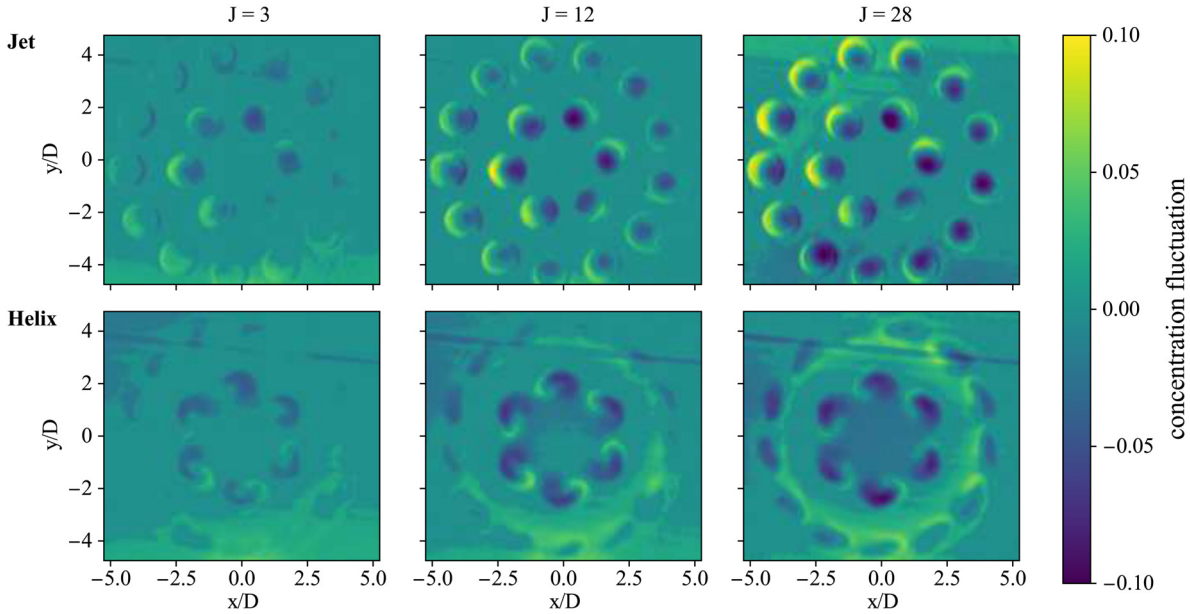


**Fig. 5 Fluctuation around the mean concentration distribution $\langle \tilde{C}(\boldsymbol{x}) \rangle$ for the jet (top row) and helix burner (bottom row) at a constant air mass flow of 0.8 $\dot{m}_{\text{nom}}$ for three different fuel momentum ratios $J$. Since the mean values are subtracted, zero equals the spatial mean. The data are clipped at ±1.5% of the mean value for a better visibility.**

part of the tube walls, oriented away from the global radial center of the burner. Since this is the case for all tubes, assembly tolerances can be ruled out as cause for this phenomenon and the nozzle positions are designed to be in the center of the tubes. Considering the higher velocity of the injected fuel jet surrogate and the increased flow volume, Bernoulli's law suggests a higher pressure for the unmixed fraction of the air surrogate flow in the tube. Since all tubes are fed with air surrogate from the shared burner plenum, the pressure of each individual tube also affects the pressure in the burner plenum which in turn would feedback to the tubes. The observed outward orientation of the fuel jets might be supported by the pressure reduction toward the burner radial center, which in turn would favor higher flow velocities at the outside. Pressure reduction toward the center is assumed to be caused by the blockage of the outer nozzles and tubes.

The second row of Fig. 5 depicts the concentration fluctuations of the helix burner, which look quite different compared to the jet burner. Note that the tubes twist clockwise around the radial center of the burner and exit the burner plate at a specific angle causing a swirling motion of the flow. All tubes show a fuel enriched region at the same location of the tube outlet. In case of the inner ring this results in forming wave-like shaped engulfed areas of concentration above and beneath the mean concentration value (best seen for the medium momentum ratio). The reason for the described concentration distributions is most likely that the fuel jets are attaching to the tube walls at the point where the twist of the tubes starts and then staying attached to this side of the wall. With an increase of the momentum ratio, the deviation from the mean value for the inner ring increases and becomes leaner. Additionally, the fuel enriched regions of the wave-like patterns are diminishing. For the outer ring, a concentration distribution is observed that is closer to the mean overall, with a slight increase in deviation as the amount of fuel increases.

Another difference between helix and jet burner arises in the center of the burners. The jet burner configuration shows a uniform concentration distribution near the global mean in the center region. In case of the helix type burner, however, the combination of the wall attached fuel and the swirling motion of the flow results in a lean concentration distribution in the center region of the burner within the inner ring.

If the fuel-rich portion of the flow were always attached to the same circumferential location of the tubes, this would most likely decrease the flashback resistance, especially if the mean concentration in this area is increased. However, the individual snapshots (not shown) recorded to calculate the mean images show that the fuel-rich shares of the jets do not remain permanently attached to the same position within the tubes, but oscillate along the circumference of the tube walls. Therefore, in Fig. 6, the temporal unmixedness factor for helix and jet burner is depicted for a momentum ratio of $J = 12$. In the upper half of the picture, the helix burner is shown and in the lower half the jet burner, separated by the white dashed line. While the absolute value of unmixedness for the outer rings is quite low with $U_t \leq 0.06$, it is increased for the inner rings. For the helix burner, the unmixedness of the inner tubes is at a low level ($U_t = 0.05$–$0.06$) in most areas and has a better mixture quality than the jet burner. However, the unmixedness is increased in the regions where the mean fuel concentration is also increased ($U_t = 0.075 - 0.085$). For the jet burner, on the other hand, the areas of high unmixedness ($U_t \geq 0.10$) are distributed along the tube walls. This leads to a very discontinuous localization of the fuel-rich share of the flow over time, which might improve flashback resistance for this type of configuration. In Atmospheric Combustion Tests section, the effects of the differences in mixture quality on the combustion characteristics are discussed.

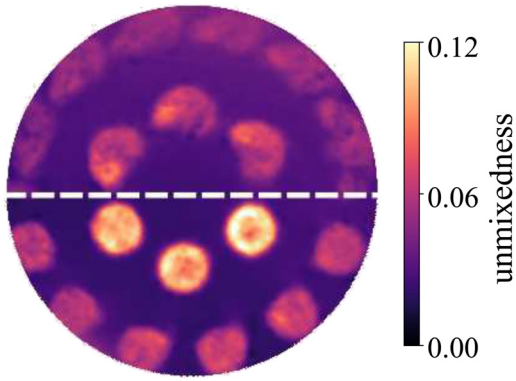


**Fig. 6 The temporal unmixedness factor of the helix burner (top half) is compared to the jet burner (bottom half) for $J = 12$, separated by the white dashed line. A value close to zero indicates a perfectly mixed fluid.**



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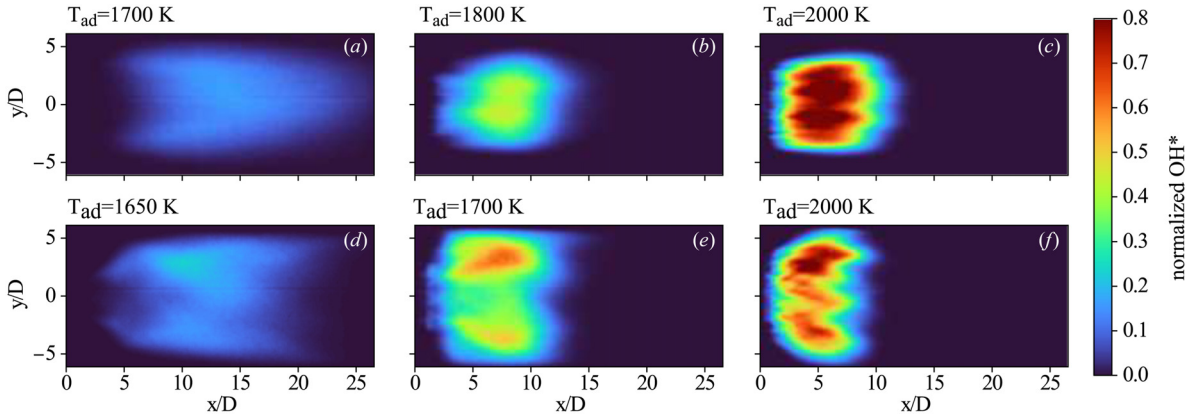


**Fig. 7 Natural gas flame OH*–chemiluminescence images of the jet burner (top row) and the helix burner (bottom row) for three different flame temperatures, respectively**

### Atmospheric Combustion Tests

*Natural Gas.* OH*–chemiluminescence images for natural gas are shown in Fig. 7. Helix and jet burner are compared at the nominal air mass flow equipped with nozzle #3. Flow direction is from left to right with the burner plate position on the left end of each image. Over the range of tested flame temperatures very different flame shapes with distinct transition points are observed (left to right). Since the flame transition occurs at different flame temperatures for each burner, the flame temperatures shown are $T_{ad} = 1700$ K, 1800 K, 2000 K and $T_{ad} = 1650$ K, 1700 K, 2000 K for the jet and the helix burner, respectively. The flame regime observed for low flame temperatures is quite narrow bounded by LBO and the first flame transition point. In this regime, at $T_{ad} \leq 1700$ K (jet burner, image *a*) and $T_{ad} \leq 1650$ K (helix burner, image *d*), the flame spreads over a large space in the combustion chamber and no individual jet flames can be observed. At this flame temperature, the laminar flame speed is only $S_L = 0.9$ m/s. Thus, the jet velocities ($u_{bulk} \approx 100\,\mathrm{m/s}$) extensively exceed the burning velocity. In case of the jet burner a flame lift of $x/D = 4$ is observed, compared to the helix burner with $x/D = 2.5$. The flame lift is defined as the distance between the burner plate and the downstream position in *x*-direction where a threshold of 1% of the maximum OH* intensity is exceeded. Reason for the smaller flame lift of the helix burner is a lower bulk velocity in *x*-direction due to the swirling motion of the flow. For all tested conditions with natural gas, the flame of the helix burner attaches to the combustion chamber walls. However, the flame shape of both burners alters fundamentally with an increasing adiabatic flame temperature. In particular, at $T_{ad} = 1700$ K (image *e*), the flow conditions allow for a reacting recirculation zone at the burner plate, as seen in swirl-stabilized burners, the so-called "M"–flames. This indicates that recirculation occurs for the flames of the helix burner, which might support flame stabilization. At $T_{ad} = 2000$ K, the flame lift decreases to $x/D = 0.6$ and $x/D = 0.35$ for the jet burner and the helix burner, respectively. This leads to a shorter mixing length and less balancing among the mixing tube streams resulting in enhanced local heat releases. The flames are now separable for the individual tubes, indicating the transition from one globally distributed flame to multiple local flames due to the increased burning velocity. The ratio of flame lift between the jet and the helix burner remains constant for all flame temperatures and is therefore controlled by the proportion of velocity in the *x*–direction. In the range of $T_{ad} = 1800$–1900 K, thermoacoustic instabilities are present which inhibit meaningful measurements. For more details on the thermoacoustic stability of the multi-tube burner, the reader is referred to Ref. [28]. For a better understanding of the flow field and the underlying mechanisms of the corresponding flame shapes, particle image velocimetry measurements have to be part of future investigations. Optical measurement data for the other fuel nozzle designs are not available due to a technical issue during the measurements.

*Hydrogen Flame.* Firing of the multi-tube burners with hydrogen is shown in Fig. 8 by means of OH*–chemiluminescence images. As for the natural gas case, nozzle #3 is installed for fuel injection. The depicted condition is at the nominal air mass flow and an adiabatic flame temperature of $T_{ad} = 1700$ K. Flow direction is from left to right with the burner plate position on the left end of each image. The hydrogen flames of the jet burner and the helix burner are rather compact with a length of $x/D = 5.9$ and $x/D = 4.9$, respectively. Compared to NG, the flame length is reduced by over 60%, and flame anchoring occurs immediately downstream of the burner plates at $x/D < 0.1$. For the helix burner, the flame is shorter and widens radially in downstream direction due to the twisted tube shape and the resulting centrifugal force. The flames of the jet burner, on the other hand, are directed slightly toward the axial centerline.

In order to analyze the flame position in more detail, Fig. 9 shows the radially integrated OH* intensity as a function of the axial position. Varied is the adiabatic flame temperature, as well as the air mass flow for the jet burner. The OH* intensity is normalized by the corresponding maximum global intensity. By varying the flame temperature, only slight modulations of the flame shape are observed, which is exemplified for an air mass flowrate of $1.2\,\dot{m}_{nom}$ in Fig. 9 (top). A maximum displacement of $x/D \approx 1$ in the upstream direction is observed over the entire range of flame temperature variation. At the same time, the position of flame anchoring and flame brush remains constant. Nevertheless, for $T_{ad} \leq 1500$ K, the sharp decrease of heat release causes a slight flame lift and thus decreases the flame anchoring, as indicated by the lower slope of the dotted line.

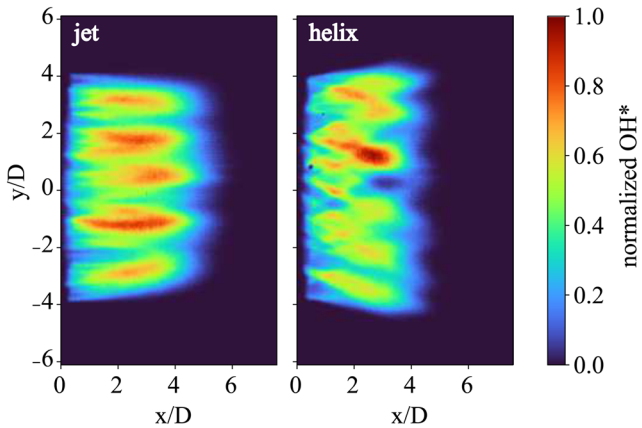


**Fig. 8 Hydrogen flame OH*–chemiluminescence images for both burner geometries at $T_{ad}$ = 1700 K**



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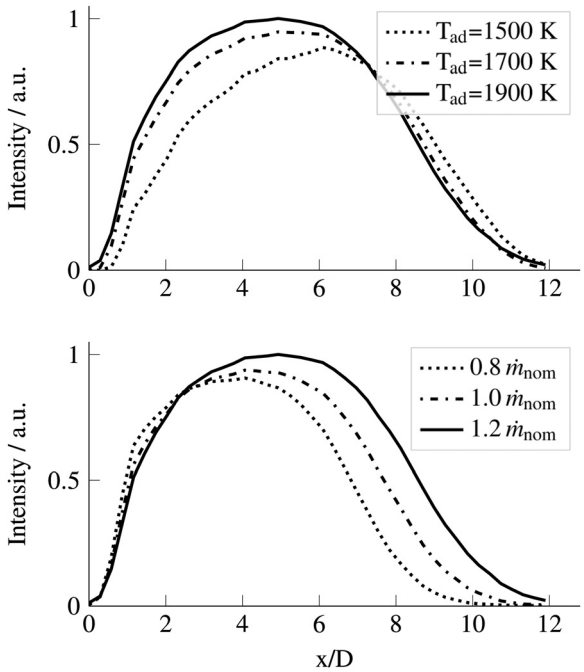


**Fig. 9 Flame positions of the jet burner identified by integrated OH* intensity in *y*-direction. Variation of the flame temperature over the range $T_{ad}$ = 1500 K, 1700 K, and 1900 K at an air mass flow of 1.2 $\dot{m}_{nom,air}$ (top) and variation of the air mass flow at $T_{ad}$ = 1900 K (bottom). The sequences are normalized by the maximum of the global intensity.**

In contrast, the variation of the air mass flow leads to a broadening of the flame in downstream direction, as shown in Fig. 9 (bottom). Depicted are 0.8–1.2 $\dot{m}_{nom}$ at a flame temperature of $T_{ad} = 1900$ K. The flame expands by $x/D \approx 2$ in the downstream direction, while flame anchoring and flame shape remain mostly unaffected.

It can be concluded that the position of the flame front is determined by the flow field. The flame stabilizes in the shear layer of the jets since the velocity gradient across the shear layer is capable to match a wide range of flame velocities without significantly altering the flame position.

*Emissions.* Figure 10 presents the measured $NO_x$ concentrations for natural gas and hydrogen combustion for both burner geometries on a dry basis normalized to 15% $O_2$. Emissions from firing natural gas and hydrogen are indicated by circle markers and square markers, respectively. The dashed lines are least square fits of the data points for a better visualization of the trend, with the blue dashed line for hydrogen and the red dashed line for natural gas. Additionally, a correction factor as proposed by Douglas et al. [29] is applied to each hydrogen measurement point. This correction allows for a comparison of natural gas and hydrogen in consideration of the actual $NO_x$ molecules produced at constant heat input. For more details on the correction of emission data, the reader is referred to Ref. [30].

Curve shapes for both fuels are quite similar on a low emission level, with under 10 ppm $NO_x$ up to a temperature of $T_{ad} = 1950$ K. Nevertheless, a steep rise of emissions already starts at a temperature of $T_{ad} = 1800$ K, which is due to the sharp decrease of mixture quality at higher fuel momentum ratios observed in the water tunnel tests. In addition, the influence of the thermal $NO_x$ pathway also increases. Based on the overall good agreement of the emission data between the different geometries, a comparable mixing quality seems to be achieved, which, however, contradicts the observations in the water tunnel with significant differences in the temporal unmixedness. Besides the qualitative character of the water tunnel measurements, one reason for this could be the neglected diffusivity that enhances the mixing quality. Although, the diffusivity of hydrogen is greater than that of natural gas, the higher flame lift and

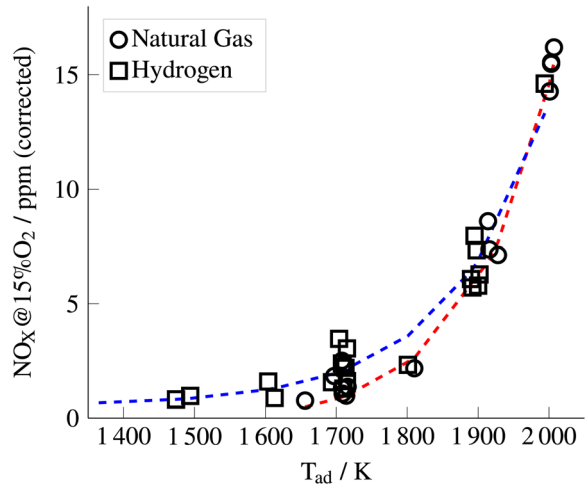


**Fig. 10 $NO_x$ emissions for hydrogen (squares, blue dashed line) and natural gas (circles, red dashed line) combustion over the adiabatic flame temperature $T_{ad}$. Hydrogen is additionally corrected with respect to exhaust gas composition. The dashed lines are least squares fits of the data points for a better visualization of the trend. (Color version online.)**

thus longer mixing length might compensate for this. Another reason for the low distinguishability of the emissions of the two burner types is the very low absolute level of the $NO_x$ emissions at atmospheric conditions. For such small numbers of ppm, a potential difference between the two burner variants might vanish within the measurement accuracy. Consequently, under high-pressure conditions, differences in $NO_x$ emissions might be detected.

Recent studies on swirl–stabilized burners, which were originally designed for natural gas combustion, suggest that switching from natural gas to hydrogen significantly increases the $NO_x$ emissions for the same gas turbine operating point [3,12]. Often, the reason for this behavior is the changed position of the reaction zone due to the increased reactivity of the hydrogen. The tested multi-tube burners show little dependence of $NO_x$ emissions on the fuel composition, indicating high potential to comply with existing emission regulations, for natural gas as well as for hydrogen.

*Impact of Burner Plate Temperature.* Obviously, the burner plate temperature has an influence on two important mechanisms: one is to ensure proper anchoring of the flame at the lower stability limit to prevent LBO, and the other is on FB propensity. The influence of the burner geometry on the base plate temperature for natural gas is depicted in Fig. 11. Shown is a variation of the main air flow with respect to the calculated adiabatic flame temperature by utilizing nozzle #1. To account for the temperature of the unburnt mixture, the measured preheat temperature of $T_2 \approx 673$ K is subtracted from the mean temperature of the inner tube ring $T_{bp}$. The inner ring is chosen because it reaches higher temperatures and thus has a greater influence on flame stability.

When firing with natural gas, the burner plate temperature is higher than the preheat temperature for both burner types. In the investigated range of $T_{ad} = 1600$–2000 K, the burner plate temperatures increase from 10 K to 20 K for the jet burner and from 40 K to 80 K for the helix burner above the preheat temperature. The effect on heat transport due to the variation of air mass flow is very limited. One reason for the higher temperatures of the helix burner might be that the flame is more compact, thus the heat flux is less distributed over the combustion chamber wall as it is for other combinations of burner geometry and fuel type. This observation is already discussed in the last section for Fig. 7. Another effect of the angled mixing tube outlets might be the resulting difference in the flow field leading to recirculations of hot combustion gases that heat the base plate as it is observable in Fig. 7(*e*). The effect of recirculations on burner material temperature was also described by [31] for a bluff body configuration. As the adiabatic flame temperature approaches the



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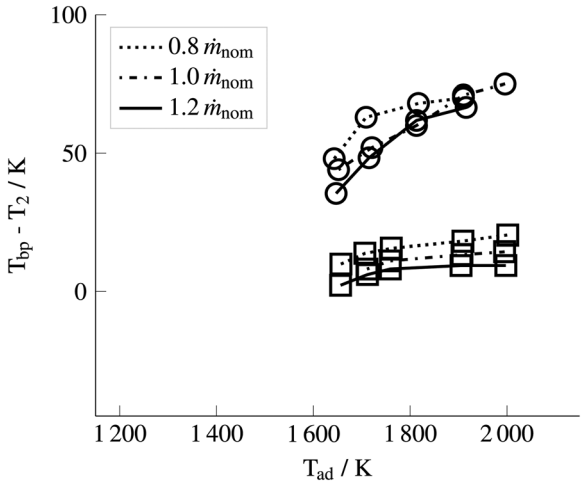


**Fig. 11 Burner plate temperatures when burning natural gas with the helix burner (circles) and jet burner (squares). Depicted are three cases of nominal air mass flow: 0.8 $\dot{m}_{nom}$ (dotted line), 1.0 $\dot{m}_{nom}$ (dashed line), and 1.2 $\dot{m}_{nom}$ (solid line).**

observed transition point of the NG flame at $T_{ad} \approx 1700$ K, the base plate temperatures of both burner types decrease significantly and thus approach their LBO limit in proximity to 1600 K. An influence of the different nozzle geometries on the base plate temperature is not observed.

By switching the fuel from natural gas to hydrogen, the base plate temperatures show a totally different behavior, as seen in Fig. 12. For the helix burner, the level of the base plate temperature at 1.0 $\dot{m}_{nom}$ is comparable to the natural gas case, but with a slightly steeper increase, especially at higher flame temperatures. For $T_{ad} = 2000$ K, the base plate reaches a temperature of 95 K above the preheat temperature. But even for very low flame temperatures, close to the LBO limit, the base plate temperature is still above the preheat temperature value. The most prominent difference for the helix type burner regarding the combustion of hydrogen and NG is the strong influence of the air mass flow. The temperature levels of the base plate for the three air mass flows tested increasingly deverge as the flame temperature increases. At $T_{ad} = 1600$ K, the temperature deviation between the tested air mass flows is already 40 K. For $T_{ad} = 1900$ K this deviation increases to 50 K between 1.0 $\dot{m}_{nom}$ and

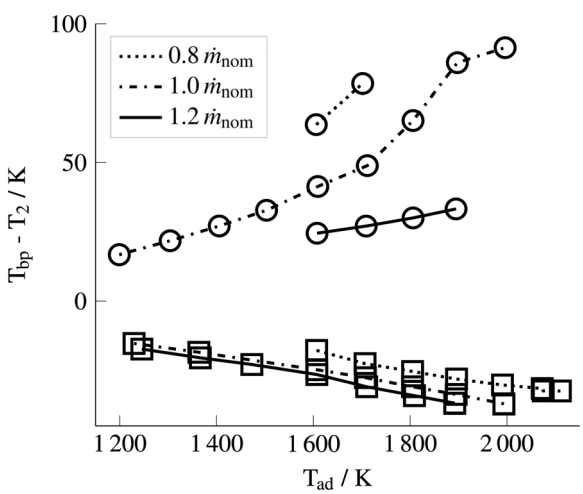


**Fig. 12 Burner plate temperatures when burning hydrogen with the helix burner (circles) and jet burner (squares). Depicted are three cases of nominal air mass flow: 0.8 $\dot{m}_{nom}$ (dotted line), 1.0 $\dot{m}_{nom}$ (dashed line), and 1.2 $\dot{m}_{nom}$ (solid line). The preheat temperature $T_2$ is subtracted from the base plate temperature $T_{bp}$.**

1.2 $\dot{m}_{nom}$ only, since for 0.8 $\dot{m}_{nom}$ FB already occurs at $T_{ad} = 1700$ K. At $T_{ad} = 1900$ K, the base plate temperature for 1.2 $\dot{m}_{nom}$ is even lower than in the NG case. Reasons for the elevated temperature of the helix burner have already been given for the natural gas case. For hydrogen, these effects are combined with the opposing effect of a lower mixture temperature of the unburnt gas. This can be shown by determining the final mixture temperature $T_{mix}$ of the fuel (subscript 1) and the air (subscript 2) under the assumption of a constant specific heat capacity $c_p$, by

$$T_{mix} = \frac{m_1 c_{p,1} T_1 + m_2 c_{p,2} T_2}{m_1 c_{p,1} + m_2 c_{p,2}} \tag{3}$$

While air mass and air temperature are equal for both fuels, the ratio

$$0.21 < \frac{m_{H2} c_{p,H2}}{m_{NG} c_{p,NG}} < 0.51 \tag{4}$$

indicates a lower mixture temperature when hydrogen is employed. This effect increases for elevated fuel mass flows. Additionally, the attachment of the injected fuel jets to the walls of the mixing tubes, which was observed in the water tunnel, also enhances the cooling effect of the hydrogen. Since no significant effect of the air mass flow on flame shape or flame position was observed in the OH*–images, this cause can be ruled out.

In contrast, the base plate temperature of the jet burner remains below the preheat temperature for all air mass flows and flame temperatures tested. It even decreases with increasing flame temperatures to a minimum of 37 K below the preheat value. This results in a temperature difference between both burners of up to 130 K at the same operating condition. The reasons can be found in the lower influence of the flame on the burner temperature combined with the cooling effect of the hydrogen that allows for higher flame temperatures without an increase of the burner plate temperature.

*Flame Stability Margins.* Limits for a reliable combustion process in this study are defined by LBO as the lower stability limit and FB as the upper stability limit.

Concerning the stability limits of the natural gas flames, no flame flashback occurred under the tested conditions due to the high flow velocity in the mixing tubes, which is supposed to ensure the stable combustion of hydrogen. The operable window for NG is nearly the same for both burners with a LBO limit in the range of $T_{ad} = 1600 - 1650$ K, independent of the air mass flow. Since the flame starts to be distributed and far lifted from the burner plate at $T_{ad} \leq 1700$ K, as observed in Figs. 7(*a*) and 7(*d*), flame anchoring becomes significantly more unstable and leads to the prompt LBO. Differences in the stability limits related to the different nozzle configurations are not identified.

The stability limits for hydrogen fuel are shown in Fig. 13. Each subplot comprises one type of fuel injection nozzle. The filled markers represent the helix burner and the empty markers represent the jet burner. The tested LBO limits are indicated with blue dashed lines, while the detected FB limits are marked with red symbols. The different level of bulk flow velocities are caused by the different air mass flow rates tested. In the test procedure, the amount of hydrogen for a constant air mass flowrate is increased until flashback occurs or the thermal limit of the test rig is reached. The observable increase of mixture bulk velocity for a specific air mass flow is caused by the increasing amount of hydrogen. This increase in bulk velocity is more pronounced for hydrogen than for natural gas due to the larger ratio between volume and thermal power. An overview on the influence of fuel amount on the bulk flow velocity is provided in Table 1.

For the helix burner, the lowest bulk velocity that is achieved for relevant flame temperatures is $u_{bulk} \approx 90$ m/s, which corresponds to an air mass flow of 0.8 $\dot{m}_{nom}$ and is independent of the utilized fuel injector nozzle. The maximum flame temperature reached for these bulk velocities is $T_{ad} = 1600\,K$. As the bulk velocity increases, the FB limits for the different fuel injection nozzles become more



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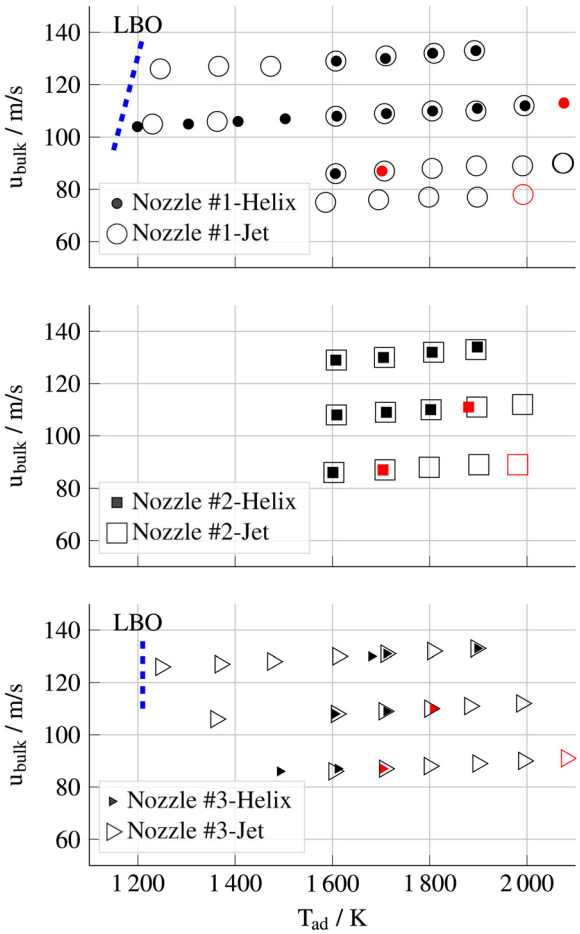


**Fig. 13 Stability limits for hydrogen combustion are grouped according to the employed types of injection nozzles. In each diagram, the filled markers show the helix burner, and the empty markers denote the jet burner. A red symbol indicates a flame flashback. The blue dashed line indicates the LBO limit, but not every configuration/condition was tested for LBO. (Color version online.)**

separable, as indicated by the red-filled markers moving to different flame temperatures in each plot. At $T_{ad} = 1800$ K nozzle #3 reaches its FB limit, closely followed by nozzle #2 at $T_{ad} = 1880$ K. For nozzle #1, the flashback limit is shifted to a significantly higher flame temperature of $T_{ad} = 2070$ K. For the highest bulk velocity tested no injector reached the FB limit in the allowed operating window of the test rig. The reason for the higher FB tendency of the helix burner was found in the elevated base plate temperatures observed in Fig. 12. Multiple investigations [32–35] described the influence of burner material temperature on FB propensity. With the increased wall temperature, the quenching distance reduces and allows for higher burning velocities in the near wall region. Due to the higher material temperatures of the inner tube ring, all observed FBs occurred at one of those mixing tubes.

The lower stability limit of the helix burner was tested for nozzle #1 only. The LBO for this configuration occurred at $T_{ad} = 1200$ K, which is 400 K below the value for the corresponding natural gas operating point. This is due to the higher chemical reactivity of hydrogen.

**Table 1 Overview of mixture bulk velocity with increasing fuel amount for the nominal air mass flow**

| Fuel | $T_{ad}$ (K) | $\rho$ (kg/m$^3$) | $u_{bulk}$ (m/s) |
|---|---|---|---|
| NG | 1647 | 0.526 | 102.7 |
| | 2003 | 0.531 | 102.8 |
| $H_2$ | 1607 | 0.492 | 108.1 |
| | 1994 | 0.478 | 111.7 |

With regard to the applied criteria for flame stability, the jet burner outperformed the helix geometry, although the LBO limit is at slightly higher flame temperatures compared to the helix type burner. For $u_{bulk} \approx 100$ m/s, which associates to the nominal air mass flow $1.0\,\dot{m}_{nom}$, all three nozzle configurations performed properly, without FB, until the thermal limit of the test rig was reached. First differences between the nozzles are found for $u_{bulk} \approx 90$ m/s, with nozzle #2 and #3 reaching their FB limits at $T_{ad} = 2000$ K and 2090 K, respectively. For nozzle #1, the bulk flow velocity had to be reduced even further to $u_{bulk} < 80$ m/s ($0.7\,\dot{m}_{nom}$) and the flame temperature increased to $T_{ad} = 2000$ K in order to achieve its FB limit for the jet burner. Thus, as with the helix burner, nozzle #1 provides the highest flashback resistance.

Therefore, for both burner geometries, the FB limit shifts to higher flame temperatures with the degree of temporal unmixedness. Although no global effect on the measured emissions could be recognized, it might be the temporal discontinuous localization of the fuel-rich share at the tube walls that prevents the occurrence of FB, even though the mean concentration is increased in the near wall region, seen in Fig. 5. The oscillation of the fuel jet of nozzle #1 is caused by flow instabilities. In contrast, nozzle #2 and #3 with the jet-in-crossflow configurations tend to increase the fuel concentration in the near wall region while exhibiting lower temporal unmixedness. As a consequence of the FB suppression, the FB mechanism will be shifted from the boundary layer to the core flow. The comparable effect of staged and homogeneous fuel-air distributions on FB was numerically investigated by Gruber et al. [36].

At the same time, the unmixedness leads to higher flow velocities in the fuel-rich regions of the mixing tube. This is assumed due to the additional momentum the coaxial jet introduces into the air flow. Based on this, Fig. 6 can also be associated with velocity fluctuations with the local velocities near the mixing tube wall continuously exceeding the bulk velocity. Both effects can sufficiently suppress backflow regions at the onset of FB. It requires further investigations to find out whether the temporal fuel concentration fluctuation or the velocity fluctuation is more crucial for FB suppression.

In order to summarize the flame stability of the multi-tube burners for hydrogen combustion, a flashback mapping for all tested conditions and configurations is presented in Fig. 14. For this purpose, the two observed main factors for flame stability in this study are related to each other. The first is the approximated burner plate temperature and the second is the chemical reactivity, which is represented by the laminar flame speed. The burner plate temperature is normalized by the auto-ignition temperature of a fuel-air mixture with $\phi = 0.45$. This equivalence ratio corresponds to the flame temperature ( $\approx$1800 K) at which most of the observed flashbacks occurred for the helix burner. The auto-ignition temperature was calculated by means of a perfectly stirred reactor model in Cantera with an estimated residence time of $\tau = 0.01$ ms. This time span is based on a bulk velocity of 100 m/s but the actual velocity near the walls where flashback is initiated is assumed to be lower, which makes it a lower estimation limit for the residence time. Therefore, the $y$-axis in Fig. 14 represents the thermally related influences on flame stability. The achievable maximum of this relation is unity, at which the fuel-air mixture would selfignite at the considered conditions. On the $x$-axis, the laminar flame speed is normalized by the bulk velocity of the unburnt fuel-air mixture, representing the ratio between chemical reactivity and flow condition that leads to a FB event following the idea of the Damköhler number. For this relation, it is not assumed that it does reach unity due to turbulent flow conditions.

All tested operating conditions and all hardware configurations for hydrogen combustion are shown in Fig. 14. Each heat map illustrates the FB propensity of a single type of fuel injector nozzle with a fade from yellow to red, where red denotes the highest FB propensity. A measured FB event is marked with a red cross. Therefore, each of the tested hardware configuration spreads out its

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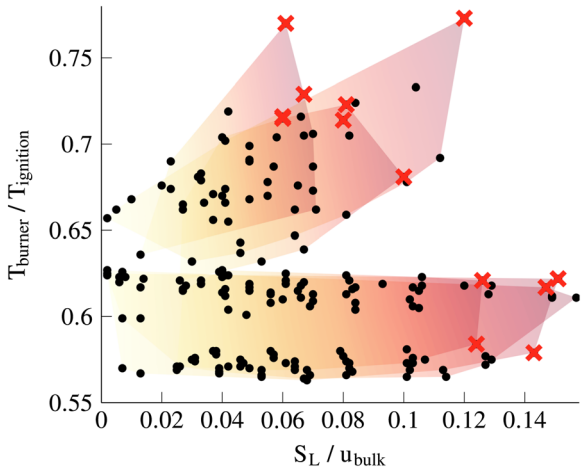


**Fig. 14 Flashback mapping for all tested conditions. Data clustering is based on the identified main factors for FB: burner plate temperature ratio and velocity ratio. The black dots mark the data points. FB events are highlighted with a red cross. Each area shows the FB propensity of a single type of fuel injector nozzle with a fade from yellow to red. Red denotes the highest FB risk. (Color version online.)**

own stability map within the plot. Two clusters of heat maps (helix and jet burner) can be identified, each with three superimposed heat maps (the nozzles). The separation between the two clusters occurs approximately at a temperature ratio of 0.63. The cluster above this temperature ratio comprises the helix burner while the cluster below belongs to the jet burner. Both areas cover very different conditions. The jet burner is completely unaffected by the temperature effects. Only the change in the experimentally adjusted preheating temperature causes the values of the temperature ratio on the $y$-axis to increase from an average value of 0.57–0.62. The increase of flame temperature leads to an increase of the laminar flame speed but without a relevant increase of the bulk velocity. This causes the areas related to the jet burner to expand horizontally to the right until FB occurs. Although the maximum value of $S_L/u_{bulk} = 0.157$ might be estimated as a small value, it must be taken into account that with $v'/S_L \leq 1$ the estimated ratio of turbulence intensity and laminar flame speed is already at a very low level. Thus, the turbulent flame speed is already much closer to the bulk velocity. However, this parameter is difficult to determine.

The second cluster, above the temperature ratio of 0.63, is related to the helix burner. The values that determine the three stability maps of this cluster completely diverge from the jet burner with increasing laminar flame speed and rise to the top right of the plot. This is due to the steep increase in burner plate temperature with increasing thermal load resulting in a FB. It is assumed that temperature ratios close to unity are actually reached in the experiments, since the tips of the thermocouples which detect the base plate temperature are not located directly at the mixing tube edges. The wider separation of the stability map boundaries for the helix burner is due to the observed deviation of base plate temperatures with increasing amounts of hydrogen for this particular geometry.

## Summary and Conclusions

The presented study investigated two geometrical variants of a multi-tube jet flame burner: the JET burner and the HELIX burner. Both are tested with three different types of fuel injectors and examined for the effects of different inlet conditions such as air preheat temperature, adiabatic flame temperature, and air bulk velocity. Pure hydrogen and pure natural gas are used as fuels in order to evaluate the fuel flexibility of the multi-tube burner. The test conditions are down-scaled from full load gas turbine operating conditions for single stage operation. PLIF measurements are conducted in a water tunnel to evaluate the resulting fuel/air mixing quality prior to combustion. Temporal unmixedness shows differences of up to 50% between the tubes of the two burner configurations. The helix burner shows a single region of enhanced unmixedness at each tube, while the regions of unmixedness for the jet burner are distributed along the circumference of each of the mixing tube inner walls. In particular, this feature appears to increase FB resistance in case of coaxial fuel injection (nozzle #1). Although qualitative differences in mixing quality were found between the burner geometries, there was no indication in the measured $NO_x$ emissions. For both burners, $NO_x$ emissions are at a low level with less than 10 ppm $NO_x$ up to a flame temperature of $T_{ad} = 1950$ K. No clear separation was found in the geometry variations either. This indicates a comparable and good mixture quality for all configurations.

In terms of flame stability margins, the differences are more pronounced between the burner geometries. For natural gas combustion, the LBO limits are at $T_{ad} \leq 1650$ K. A flame transition point is observed at $T_{ad} \leq 1700$ K, where the flame shape changes from a broad distribution over the combustion chamber to a compact flame with increasing temperature. Since the burner has been designed for pure hydrogen operation, the FB limits are not reached for natural gas within the feasible operating conditions.

With hydrogen as fuel, the base plate temperatures of the burner geometries differ greatly. At the same operating conditions, the temperature difference of the base plates can be up to 130 K. This greatly reduces the FB resistance of the helix burner while only slightly increasing the LBO limit. In addition, the FB limit is affected by the type of fuel injection nozzle, with the FB resistance increased for the coaxial fuel injection at nozzle #1.

It is concluded that the temperature of the burner base plate is important for stability margins, especially for hydrogen fuel. But also a higher unmixedness can enhance FB resistance of a multi-tube jet flame burner.

## Acknowledgment

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## Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

## Nomenclature

$C(x)$ = concentration
$\tilde{C}(x)$ = concentration fluctuation
FB = flashback
$H_2$ = hydrogen
I = intensity
J = momentum ratio
LBO = lean blow-off
$\dot{m}$ = mass flow
NG = natural gas
$NO_x$ = nitrogen oxides
PLIF = planar laser-induced fluorescence
$S_L$ = laminar burning velocity
$U_t$ = temporal unmixedness



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$\lambda$ = wavelength
$\sigma$ = standard deviation

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